Security+ SY0-701 Ports and Protocols - Corrected

Security+ SY0-701 Ports & Protocols — Corrected Reference

Reconciled against CompTIA's official SY0-701 objectives list (Cyberkraft training reference). Organized by OSI layer, matching how the exam objectives group them.

Layer 7 — Application

Protocol	Port	TCP/UDP	Use
FTP	20 / 21	TCP	Port 21 = control, Port 20 = data transfer
SSH	22	TCP	Encrypted remote connection
SFTP	22	TCP	Separate protocol from FTP; uses SSH to encrypt file transfers
SMTP	25	TCP	Sends outgoing mail, client to mail server
TACACS+	49	TCP	Cisco proprietary AAA protocol
DNS	53	UDP	Associates IP addresses with domain names
DHCP	67 / 68	UDP	Assigns local private IP addresses
HTTP	80	TCP	Standard web traffic
Kerberos	88	TCP/UDP	Network authentication; primarily UDP, can use TCP
POP3	110	TCP	One-way email retrieval (client reads from server)
NTP	123	UDP	Synchronizes timekeeping across a network
SMB (legacy/NetBIOS)	137 / 138	UDP	Name resolution (137) and datagram transfer (138)
IMAP	143	TCP	Two-way email communication (unlike POP)
SNMP	161 / 162	UDP	Monitors/manages network devices
LDAP	389	UDP	Manages/communicates with directories
HTTPS	443	TCP	Secure HTTP, uses TLS
SSTP	443	TCP	Microsoft's secure VPN tunneling protocol, uses TLS; replaces insecure PPTP/L2TP-IPsec
SMB (modern)	445	TCP	Direct remote server access (replaces legacy 137-139 usage)
IPsec (ISAKMP)	500	UDP	Security association/key management for IPsec
SMTPS	587	TCP	Secure SMTP, uses TLS
LDAPS	636	TCP	Secure LDAP, uses TLS
FTPS	989 / 990	TCP	Secure FTP using TLS; distinct from FTP's 20/21
IMAPS	993	TCP	Secure IMAP, uses TLS
POP3S	995	TCP	Secure POP3, uses TLS
RADIUS	1812 / 1813	UDP	AAA for network services
RDP	3389	TCP	Windows remote desktop connections
Diameter	3868	TCP	Modern upgrade to RADIUS
SRTP	5004	UDP	Secure streaming audio/video (replaces RTP)

Layer 4 — Transport

Protocol	Port	TCP/UDP	Use
TCP	N/A	TCP	Reliable transmission with error checking

UDP	N/A	UDP	Fast, best-effort transmission, no error checking
PPTP	1723	TCP	Deprecated VPN protocol, based on PPP

Layer 2 — Data Link

Protocol	Port	TCP/UDP	Use
L2TP	1701	UDP	Point-to-point VPN tunneling; needs IPsec for encryption; extension of PPTP
PPTP	1723	(see note)	Same protocol as above — appears at both layers in official material due to its control (TCP)/data (GRE) split

Encrypted vs. Unencrypted Pairs — memorize these together

Unencrypted	Port	Encrypted version	Port
FTP	20/21	FTPS	989/990
Telnet	23	SSH	22
HTTP	80	HTTPS	443
POP3	110	POP3S	995
IMAP	143	IMAPS	993
SMTP	25	SMTPS	587
LDAP	389	LDAPS	636

Corrections from earlier version of this guide

If you saw an earlier version of this reference from me, these are the fixes: - **SMTPS** → exam tests **587**, not 465 - **SMB** → has TWO entries: legacy 137/138 (UDP) and modern 445 (TCP) — not just 445 - **IPsec** → only **500 (UDP)** is on the core objectives list; 4500 (NAT-T) is real-world but not the tested port - **Added:** SSTP (443/TCP), FTPS (989/990/TCP), SRTP (5004/UDP) — these were missing before